Language and Compilers

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Sheet (2)

Weeks 2 & 3

1- Define the terms (syntax analysis ,CFG)
2- Given the following simple English grammar

S → NP VP END

NP → DET N | DET ADJ N

VP → V NP | V PREP NP | VP PREP NP (PREP : Preposition)

END → DOT | SEMICOLON

DET → A |a | the | The

ADJ → beautiful | young |nice

N → boy | boys | girl | girls | dress | ball |book | table

V → kicked | appeared | played| put

PREP→on | in | at | by | over

DOT → .

SEMICOLON → ;

Show the parse tree for each of the following sentences:

- A beautiful girl appeared in a nice hat.
- A young boy kicked the ball.
- The boy put the book on the table.

3) Let a CFG {N,T,P,S} be

N = {S}, T = {a, b}, Starting symbol = S, P as following :

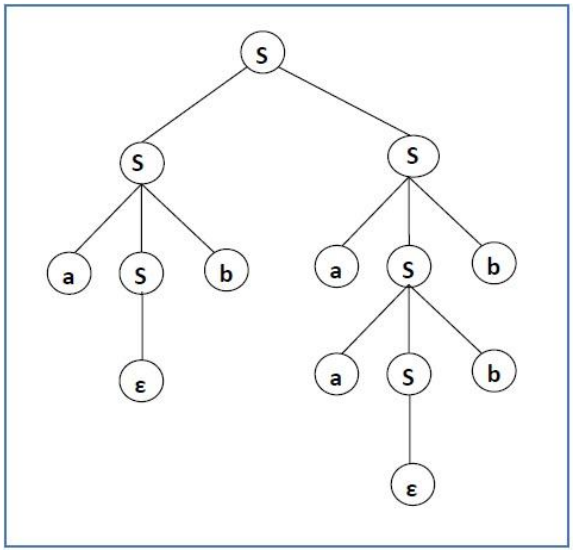
S → SS | aSb | ε

Show how to derive the string “abaabb” from the above CFG is

Sol :

S → SS →aSbS→abS→abaSb→abaaSbb→abaabb

Show the derivation tree ?



4)define the following terms (lexeme , token, pattern)

A lexeme is a sequence of characters in the source program that matches the pattern for a token and is identified by the lexical analyzer as an instance of that token.

A token is a pair consisting of a token name and an optional attribute value. The token name is an abstract symbol representing a kind of lexical unit, e.g., a particular keyword, or a sequence of input characters denoting an identifier.

A pattern is a description of the form that the lexemes of a token may take.

5) give a description of the tasks of lexical analyzer.

the main task of the lexical analyzer is

- **To read the input characters of the source program, group them into lexemes, and produce as output a sequence of tokens for each lexeme in the source program. The stream of tokens is sent to the parser for syntax analysis.**
- **To strip out whitespace (blank, newline , tabs) and comments.**
- **Also it can count the number of newline characters, so it give the line number corresponding to each error message.**

6) Define the terms (alphabet, language, string)

An *alphabet* is any finite set of symbols. Typical examples of symbols are letters , digits, and punctuation.

***A language* is any countable set of strings over some fixed alphabet.**

A *string* over an alphabet is a finite sequence of symbols drawn from that alphabet. In language theory, the terms "sentence" and "word" are often used as synonyms for "string." The length of a string s, usually written as |S| , is the number of occurrences of symbols in s. The empty string, denoted ϵ, is the string of length zero.

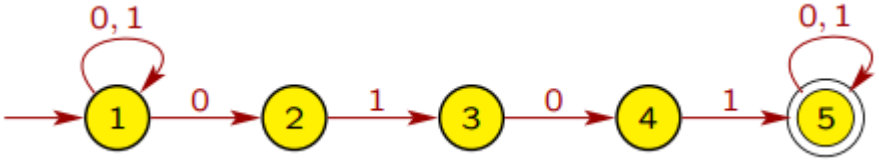
7)show the CFG with alphabet {0,1} such that it generates the following language {0n 1n+1: n≥=1}

Sol :

S→ A1

A→ 0A1 | λ

8)describe the language accepted by finite automata whose transition diagrams are given in the following figure:



Sol : (0|1)*0101(0|1)*

Give the regular expressions that represent the following strings over the alphabet {a,b,c}:

- All strings containing even number of b's
- **solution** : (a|c)* (bb)* (a|c)* (bb)* (a|c)*
- all strings containing exactly one a.

  Solution (b | c)* a(b | c)*

(b) All strings that contain an even number of b's.

$$a^*(ba^*ba^*)^*$$

(c) All strings which do not contain the substring ba.

$$a^*b^*$$

Example 3.4: Let $\Sigma = \{a, b\}$.

1. The regular expression $\mathbf{a}|\mathbf{b}$ denotes the language $\{a, b\}$.

2. $(\mathbf{a}|\mathbf{b})(\mathbf{a}|\mathbf{b})$ denotes $\{aa, ab, ba, bb\}$, the language of all strings of length two over the alphabet Σ. Another regular expression for the same language is $\mathbf{aa}|\mathbf{ab}|\mathbf{ba}|\mathbf{bb}$.

3. $\mathbf{a}^*$ denotes the language consisting of all strings of zero or more a's, that is, $\{\epsilon, a, aa, aaa, \ldots\}$.

4. $(\mathbf{a}|\mathbf{b})^*$ denotes the set of all strings consisting of zero or more instances of a or b, that is, all strings of a's and b's: $\{\epsilon, a, b, aa, ab, ba, bb, aaa, \ldots\}$. Another regular expression for the same language is $(\mathbf{a}^*\mathbf{b}^*)^*$.

5. $\mathbf{a}|\mathbf{a}^*\mathbf{b}$ denotes the language $\{a, b, ab, aab, aaab, \ldots\}$, that is, the string a and all strings consisting of zero or more a's and ending in b.

$$
\begin{aligned}
R &\mapsto XRX \mid S \\
S &\mapsto aTb \mid bTa \\
T &\mapsto XTX \mid X \mid \epsilon \\
X &\mapsto a \mid b
\end{aligned}
$$

a. The variables of G are $\{R, X, S, T\}$. The terminals of G are $\{a, b\}$. The start variable is R.

b. Strings in $L(G)$ are ab, ba, and aaaabbaaa.

c. Strings not in $L(G)$ are aaa, aba, and bb.

4.3 Writing a Grammar : Regular Expressions vs. CFG's

- Can rewrite a reg. exp. as a CFG: e.g., $(a \mid b)^*abb$:

$$A_0 \rightarrow aA_0 \mid bA_0 \mid aA_1$$
$$A_1 \rightarrow bA_2$$
$$A_2 \rightarrow bA_3$$
$$A_3 \rightarrow \in$$

1. Give a context-free grammar (CFG) for each of the following languages over the alphabet $\Sigma = \{a, b\}$:

   (a) All strings in the language $L : \{a^n b^m a^{2n} | n, m \geq 0\}$

$$\begin{aligned} S &\rightarrow aSaa \mid B \\ B &\rightarrow bB \mid \epsilon \end{aligned}$$

   (b) All nonempty strings that start and end with the same symbol.

$$\begin{aligned} S &\rightarrow aXa \mid bXb \mid a \mid b \\ X &\rightarrow aX \mid bX \mid \epsilon \end{aligned}$$

   (c) All strings with more a's than b's.

$$\begin{aligned} S &\rightarrow Aa \mid MS \mid SMA \\ A &\rightarrow Aa \mid \epsilon \\ M &\rightarrow \epsilon \mid MM \mid bMa \mid aMb \end{aligned}$$